CO3- Assessment Tool 1- Case Study

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Case Study 1 – Employee Salaries

Problem Statement

A company records the annual salaries (in ₹ lakhs) of 9 employees as follows:

4, 5, 5, 6, 7, 8, 8, 9, 30

The objective is to analyze the salary distribution using measures of central tendency and dispersion, identify any outliers, and determine the most appropriate measure that represents the typical employee salary.

Given Data

4, 5, 5, 6, 7, 8, 8, 9, 30

Number of observations:

$$n = 9$$

Sum of observations:

$$x = 4 + 5 + 5 + 6 + 7 + 8 + 8 + 9 + 30 = 82$$

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1. Calculate the Mean, Median and Mode

(A) Mean (Arithmetic Mean)

Definition

The arithmetic mean is the average of all observations. It is calculated by dividing the sum of all observations by the total number of observations.

Formula

$$\bar{x} = \frac{\sum x}{n}$$

where

- $\bar{x}$= Mean

- $\sum x$= Sum of observations
- n= Number of observations

Calculation

$$\bar{x} = \frac{82}{9}$$

$$\boxed{\bar{x}}\boxed{= 9.11}$$

Interpretation

The average salary of employees is approximately ₹9.11 lakhs. However, this average is influenced by the unusually high salary of ₹30 lakhs.

(B) Median

Definition

The median is the middle value of a data set arranged in ascending order. It divides the data into two equal halves.

Formula

For an odd number of observations,

$$\text{Median Position} = \frac{n+1}{2}$$

Calculation

Since

$$n = 9$$

Median position

$$= \frac{9+1}{2} = 5$$

The fifth observation is

$$7$$

Therefore,

$$\boxed{\text{Median} = 7 \text{ lakhs}}$$

Interpretation

Half of the employees earn less than ₹7 lakhs, while the remaining half earn more than ₹7 lakhs.

(C) Mode

Definition

The mode is the value that occurs most frequently in the data set.

Frequency Distribution

Salary (₹ Lakhs)	Frequency
4	1
5	2
6	1
7	1
8	2
9	1
30	1

The highest frequency is 2.

Hence,

$$\boxed{\text{Mode} = 5 \text{ and } 8}$$

The distribution is bimodal because there are two modes.

Interpretation

The most common salaries in the company are ₹5 lakhs and ₹8 lakhs.

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2. Calculate the Range

Definition

The range measures the spread of the data by finding the difference between the largest and smallest values.

Formula

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

Calculation

Maximum salary

$$= 30$$

Minimum salary

$$= 4$$

Therefore,

$$\text{Range} = 30 - 4 = 26$$

$$\boxed{\text{Range} = 26 \text{ lakhs}}$$

Interpretation

There is a difference of ₹26 lakhs between the highest-paid and lowest-paid employee, indicating a large spread in salaries.

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3. Calculate the Variance and Standard Deviation

Variance

Definition

Variance measures how far each observation is from the mean. A larger variance indicates greater variability in the data.

Formula (Population Variance)

$$\sigma = \frac{(x - \bar{x})}{n}$$

where

- σ =Variance
- $\bar{x}$ = 9.11

Step 1: Calculate Deviations

Salary (x)	$x - \bar{x}$	$(x - \bar{x})$
4	-5.11	26.11
5	-4.11	16.90
5	-4.11	16.90
Salary (x)	$x - \bar{x}$	$(x - \bar{x})$
6	-3.11	9.68
7	-2.11	4.46
8	-1.11	1.23
8	-1.11	1.23
9	-0.11	0.01
30	20.89	436.23

$$(x - \bar{x}) = 512.75$$

Step 2: Variance

$$\sigma = \frac{512.75}{9}$$

$$\boxed{\sigma = 56.97}$$

Standard Deviation

Definition

Standard deviation is the square root of variance. It indicates how much the values typically deviate from the mean.

Formula

$$\sigma = \sqrt{\sigma}$$

Calculation

$$\sigma = \sqrt{56.97}$$

$$\boxed{\sigma = 7.55}$$

Interpretation

A standard deviation of 7.55 lakhs indicates considerable variation in employee salaries, mainly due to the extremely high salary of ₹30 lakhs.

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4. Identify the Outlier

The Interquartile Range (IQR) Method is used.

Step 1: Find Quartiles

Data:

$$4,\ 5,\ 5,\ 6,\ 7,\ 8,\ 8,\ 9,\ 30$$

Median (Q_2)

$$Q \quad = 7$$

Lower half

$$4,\ 5,\ 5,\ 6$$
$$Q \quad = \frac{5+5}{2} = 5$$

Upper half

$$8,\ 8,\ 9,\ 30$$
$$Q \quad = \frac{8+9}{2} = 8.5$$

Step 2: Calculate IQR

Formula

$$IQR = Q \quad - Q$$
$$= 8.5 - 5$$
$$= 3.5$$

Step 3: Calculate Lower and Upper Limits

Lower Limit

$$
\begin{aligned}
&Q \quad - 1.5(IQR) \\
&= 5 - 1.5(3.5) \\
&= 5 - 5.25 \\
&= -0.25
\end{aligned}
$$

Upper Limit

$$
\begin{aligned}
&Q \quad + 1.5(IQR) \\
&= 8.5 + 5.25 \\
&= 13.75
\end{aligned}
$$

Any value above 13.75 is considered an outlier.

Since

$$30 > 13.75$$

$$\boxed{\text{30 lakhs is an outlier}}$$

Interpretation

The salary of ₹30 lakhs is significantly higher than the rest of the salaries and does not represent the general salary pattern of the employees.

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5. Which Measure of Central Tendency Best Represents the Typical Salary? Explain

Among the three measures of central tendency:

- Mean = 9.11 lakhs
- Median = 7 lakhs
- Mode = 5 and 8 lakhs

The median is the most appropriate measure.

Reasons

- The salary of ₹30 lakhs is an extreme value (outlier).
- This outlier pulls the mean upward from the actual central value.
- The median remains unaffected by extreme observations.
- Therefore, the median better represents the salary earned by a typical employee.

Hence,

$$\boxed{\text{Median (₹7 lakhs) is the best measure of central tendency.}}$$

Conclusion

This case study demonstrates how different statistical measures describe a dataset. Although the mean salary is ₹9.11 lakhs, it is inflated by the ₹30 lakh outlier. The median salary of ₹7 lakhs provides a more realistic representation of the typical employee's earnings. The high range (26 lakhs), variance (56.97), and standard deviation (7.55 lakhs) indicate substantial variation in salaries. The IQR method confirms that ₹30 lakhs is an outlier, showing why the median is preferred over the mean for this dataset. This analysis highlights the importance of choosing appropriate statistical measures when data contains extreme values.

Case Study 2 – Website Response Time

Problem Statement

A data scientist records the response time (in milliseconds) of a website for 10 requests as follows:

$$120,\ 125,\ 128,\ 130,\ 132,\ 135,\ 140,\ 145,\ 150,\ 300$$

The objective is to analyze the website response times using measures of central tendency, dispersion, and skewness, and determine whether the mean or median better represents the typical response time.

Given Data

$$120,\ 125,\ 128,\ 130,\ 132,\ 135,\ 140,\ 145,\ 150,\ 300$$

Number of observations:

$$n = 10$$

Sum of observations:

$$120 + 125 + 128 + 130 + 132 + 135 + 140 + 145 + 150 + 300 = 1505$$

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1. Calculate the Mean and Median

(A) Mean (Arithmetic Mean)

Definition

The arithmetic mean is the average of all observations.

Formula

$$\bar{x} = \frac{\sum x}{n}$$

where

- $\bar{x}$= Mean
- $\sum x$= Sum of observations
- n= Number of observations

Calculation

$$\bar{x} = \frac{1505}{10}$$

$$\boxed{\bar{x}} \boxed{= 150.5 \text{ ms}}$$

Interpretation

The average website response time is 150.5 ms. However, this average is affected by the unusually high response time of 300 ms.

(B) Median

Definition

The median is the middle value of the ordered data. For an even number of observations, it is the average of the two middle values.

Formula

$$\text{Median} = \frac{\frac{n}{2} \text{ value} + \frac{n}{2} + 1 \text{ value}}{2}$$

Calculation

Since

$$n = 10$$

The middle two values are

- 5th value = 132

- 6th value = 135

Therefore,

$$\text{Median} = \frac{132 + 135}{2}$$
$$= \frac{267}{2}$$
$$\boxed{133.5 \text{ ms}}$$

Interpretation

Half of the response times are below 133.5 ms, and half are above 133.5 ms.

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2. Calculate the Range

Definition

The range measures the spread of the data by finding the difference between the largest and smallest values.

Formula

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

Calculation

Maximum response time

$$= 300 \text{ ms}$$

Minimum response time

$$= 120 \text{ ms}$$

Therefore,

$$\text{Range} = 300 - 120$$
$$\boxed{\text{Range} = 180 \text{ ms}}$$

Interpretation

The website response times vary by 180 ms, indicating a large spread due to one unusually high response time.

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3. Calculate the Variance and Standard Deviation

Since all observations are included, use the population variance formula.

Variance

Definition

Variance measures how much the response times vary from the mean.

Formula

$$\sigma = \frac{(x - \bar{x})}{n}$$

where

- $\bar{x} = 150.5$

Step 1: Calculate Deviations

Response Time (x)	$x - \bar{x}$	$(x - \bar{x})$
120	-30.5	930.25
125	-25.5	650.25
128	-22.5	506.25
130	-20.5	420.25
132	-18.5	342.25
135	-15.5	240.25
140	-10.5	110.25
145	-5.5	30.25
150	-0.5	0.25
300	149.5	22350.25

$$(x - \bar{x}) = 25580.50$$

Step 2: Variance

$$\sigma = \frac{25580.50}{10}$$

$$\boxed{\sigma = 2558.05 \text{ ms}}$$

Standard Deviation

Definition

Standard deviation is the square root of variance and represents the average deviation from the mean.

Formula

$$\sigma = \sqrt{\sigma}$$

Calculation

$$\sigma = \sqrt{2558.05}$$

$$\boxed{\sigma = 50.58 \text{ ms}}$$

Interpretation

The response times typically vary by about 50.58 ms from the average. The large standard deviation is mainly caused by the 300 ms response time.

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4. Determine Whether the Data is Right-Skewed or Left-Skewed

Definition

A distribution is:

- Right-skewed (positively skewed) when there are one or more unusually large values.
- Left-skewed (negatively skewed) when there are one or more unusually small values.

Observation

- Mean = 150.5 ms
- Median = 133.5 ms

Since

$$\boxed{\text{Mean} > \text{Median}}$$

and the value 300 ms is much larger than the other observations, the distribution has a long tail on the right side.

Therefore,

$$\boxed{\text{The data is positively (right) skewed.}}$$

Interpretation

Most response times lie between 120 ms and 150 ms, while one unusually high response time (300 ms) pulls the average to the right.

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5. Why is the Median More Useful Than the Mean?

Explanation

Although the mean response time is 150.5 ms, it is increased by the extreme value of 300 ms.

The median (133.5 ms) is not affected by this outlier and therefore provides a better representation of the typical website response time.

Reasons

- The response time of 300 ms is an extreme value.
- The mean is sensitive to extreme observations.
- The median remains stable even when outliers are present.
- Therefore, the median better represents the normal performance of the website.

Hence,

$$\boxed{\text{The median is a more reliable measure of central tendency for this dataset.}}$$

Conclusion

This case study shows how an outlier can influence statistical measures. The mean response time (150.5 ms) is higher than the median (133.5 ms) because of the unusually slow 300 ms response. The range (180 ms), variance (2558.05 ms^2), and standard deviation (50.58 ms) indicate significant variability in the response times. Since the distribution is positively (right) skewed, the median is the most appropriate measure of central tendency because it better reflects the typical website response time without being affected by the extreme value.

Final Answers

Measure	Answer
Mean	150.5 ms
Median	133.5 ms
Range	180 ms
Variance	2558.05 ms^2
Standard Deviation	50.58 ms
Distribution	Right-skewed (Positively Skewed)
Best Measure of Central Tendency	Median (133.5 ms), because it is not affected by the extreme response time of 300 ms.

Case Study 3 – Student Performance

Problem Statement

The marks obtained by 12 students in a Data Science test are given below:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

The objective is to analyze the student performance using quartiles, interquartile range (IQR), outlier detection, mean, and median, and explain how an outlier affects the mean.

Given Data

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

Number of observations:

$$n = 12$$

Sum of observations:

$$42 + 45 + 48 + 50 + 52 + 52 + 55 + 58 + 60 + 62 + 65 + 95 = 684$$

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1. Find Q_1, Q_2 and Q_3

Definition

Quartiles divide an ordered dataset into four equal parts.

- Q_1 (First Quartile) – 25% of the data lies below this value.
- Q_2 (Second Quartile) – Median of the dataset.
- Q_3 (Third Quartile) – 75% of the data lies below this value.

Step 1: Arrange the Data

The data is already arranged in ascending order.

$$42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95$$

Step 2: Calculate Q_2 (Median)

Formula (Even Number of Observations)

$$Q = \frac{\frac{n}{2} \text{ value} + \frac{n}{2} + 1 \text{ value}}{2}$$

Since

$$n = 12$$

Middle two values are

- 6th value = 52
- 7th value = 55

Therefore,

$$Q = \frac{52 + 55}{2}$$

$$= \frac{107}{2}$$

$$\boxed{Q \;= 53.5}$$

Step 3: Calculate Q_1

Lower half of the data:

$$42,\ 45,\ 48,\ 50,\ 52,\ 52$$

Middle two values:

48 and 50

$$Q \;= \frac{48 + 50}{2}$$

$$= \frac{98}{2}$$

$$\boxed{Q \;= 49}$$

Step 4: Calculate Q_3

Upper half:

$$55,\ 58,\ 60,\ 62,\ 65,\ 95$$

Middle two values:

60 and 62

$$Q \;= \frac{60 + 62}{2}$$

$$= \frac{122}{2}$$

$$\boxed{Q \;= 61}$$

---

2. Calculate the Interquartile Range (IQR)

Definition

The Interquartile Range (IQR) measures the spread of the middle 50% of the data.

Formula

$$IQR = Q \;\; - Q$$

Calculation

$$IQR = 61 - 49$$
$$\boxed{IQR = 12}$$

Interpretation

The middle 50% of student marks are spread across 12 marks, indicating moderate variability among most students.

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3. Use the 1.5 × IQR Rule to Identify Any Outliers

Definition

An observation is considered an outlier if it is:

- Less than

$$Q \quad - 1.5(IQR)$$

or greater than

$$Q \quad + 1.5(IQR)$$

Step 1: Calculate Lower Limit

Formula

$$Q \quad - 1.5(IQR)$$

Calculation

$$49 - 1.5(12)$$
$$49 - 18$$
$$\boxed{31}$$

Step 2: Calculate Upper Limit

Formula

$$Q \quad + 1.5(IQR)$$

Calculation

$$61 + 1.5(12)$$
$$61 + 18$$
$$\boxed{79}$$

Step 3: Identify Outliers

Any value less than 31 or greater than 79 is an outlier.

The only value greater than 79 is

$$95$$

Therefore,

$$\boxed{95 \text{ is an outlier}}$$

Interpretation

The mark 95 is significantly higher than the rest of the class and is considered an outlier.

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4. Find the Mean and Median

(A) Mean

Definition

The mean is the average of all observations.

Formula

$$\bar{x} = \frac{\sum x}{n}$$

Calculation

$$\bar{x} = \frac{684}{12}$$
$$\boxed{\bar{x}} \boxed{= 57}$$

Interpretation

The average mark scored by the students is 57.

(B) Median

The median has already been calculated as

$$\boxed{53.5}$$

Interpretation

Half of the students scored below 53.5 marks, while the other half scored above 53.5 marks.

---

5. Explain How the Outlier Affects the Mean

Explanation

The score 95 is much higher than the rest of the marks and acts as an outlier.

Because the mean uses every value in the dataset, the unusually high score increases the average mark.

As a result,

- Mean = 57
- Median = 53.5

Since

$$57 > 53.5$$

the mean is pulled upward by the outlier.

The median, however, depends only on the middle values and is not significantly affected by the extreme score.

Therefore, the median provides a better representation of the typical student performance.

Conclusion

This case study demonstrates the importance of quartiles, IQR, and outlier detection in data analysis. The quartiles are $Q_1 = 49$, $Q_2 = 53.5$, and $Q_3 = 61$, with an IQR of 12. Applying the $1.5 \times$ IQR rule identifies 95 as an outlier. The mean mark is 57, while the median is 53.5. The outlier increases the mean, making it higher than the median. Therefore, the median better represents the typical student's performance because it is less influenced by extreme values.

Final Answers

Measure	Answer
Q_1	49
Q_2 (Median)	53.5
Q_3	61
IQR	12
Lower Limit	31
Upper Limit	79
Outlier	95
Mean	57
Median	53.5
Effect of Outlier	The outlier (95) increases the mean, while the median remains relatively unaffected. Therefore, the median better represents the typical student's performance.

Case Study 4 – E-Commerce Orders

Problem Statement

An online store records the number of orders received over 10 days as follows:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

The objective is to analyze the order data using measures of central tendency, dispersion, and coefficient of variation, and interpret what the standard deviation indicates about the variability of daily orders.

Given Data

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

Number of observations:

$$n = 10$$

Sum of observations:

$$18 + 20 + 22 + 22 + 24 + 25 + 26 + 28 + 30 + 35 = 250$$

---

1. Calculate the Mean, Median and Mode

(A) Mean (Arithmetic Mean)

Definition

The arithmetic mean is the average number of orders received over the 10 days.

Formula

$$\bar{x} = \frac{\sum x}{n}$$

where

- $\bar{x}$= Mean
- $\sum x$= Sum of observations
- n= Number of observations

Calculation

$$\bar{x} = \frac{250}{10}$$

$$\boxed{\bar{x}} \boxed{= 25}$$

Interpretation

On average, the online store received 25 orders per day.

(B) Median

Definition

The median is the middle value of the ordered data. Since there are an even number of observations, it is the average of the two middle values.

Formula

$$\text{Median} = \frac{\left(\frac{n}{2}\right)\text{value} + \left(\frac{n}{2}+1\right)\text{value}}{2}$$

Calculation

The ordered data is already arranged.

The middle two values are:

- 5th value = 24
- 6th value = 25

Therefore,

$$\text{Median} = \frac{24 + 25}{2}$$

$$= \frac{49}{2}$$

$$\boxed{\text{Median} = 24.5}$$

Interpretation

Half of the days recorded less than 24.5 orders, while the other half recorded more than 24.5 orders.

(C) Mode

Definition

The mode is the value that occurs most frequently.

Frequency Table

Orders	Frequency
18	1
20	1
22	2
24	1
25	1
26	1

28	1
30	1
35	1

The value 22 appears twice, while all others appear only once.

Therefore,

$$\boxed{\text{Mode} = 22}$$

Interpretation

The most frequently recorded number of daily orders is 22.

---

2. Calculate the Variance and Standard Deviation

Since all 10 days are considered, use the population variance formula.

Variance

Definition

Variance measures how much the daily orders vary from the mean.

Formula

$$\sigma = \frac{(x - \bar{x})}{n}$$

where

- $\bar{x} = 25$

Step 1: Calculate Deviations

Orders (x)	$x - \bar{x}$	$(x - \bar{x})$
18	-7	49
20	-5	25
22	-3	9
22	-3	9
24	-1	1

25	0	0
26	1	1
28	3	9
30	5	25
35	10	100

$$(x - \bar{x}) = 228$$

Step 2: Variance

$$\sigma = \frac{228}{10}$$

$$\boxed{\sigma = 22.8}$$

Standard Deviation

Definition

Standard deviation is the square root of the variance. It shows the average deviation of observations from the mean.

Formula

$$\sigma = \sqrt{\sigma}$$

Calculation

$$\sigma = \sqrt{22.8}$$

$$\boxed{\sigma = 4.77}$$

Interpretation

The daily orders differ from the average by approximately 4.77 orders.

---

3. Calculate the Coefficient of Variation (CV)

Definition

The coefficient of variation measures the relative variability of the data with respect to the mean.

Formula

$$CV = \frac{\sigma}{\bar{x}} \times 100$$

where

- $\sigma = 4.77$
- $\bar{x} = 25$

Calculation

$$CV = \frac{4.77}{25} \times 100$$
$$= 0.1908 \times 100$$
$$\boxed{CV = 19.08\%}$$

Interpretation

The daily number of orders varies by approximately 19.08% of the average number of orders, indicating relatively stable order volumes.

---

4. What Does the Standard Deviation Tell the Company?

Explanation

The standard deviation of 4.77 orders indicates how much the daily orders fluctuate around the average of 25 orders.

- A small standard deviation means daily orders remain close to the average.
- A large standard deviation means there is significant fluctuation in daily orders.

Since the standard deviation is only about 4.77 orders, the store experiences fairly consistent daily demand with only moderate changes from day to day.

This information helps the company:

- Plan inventory levels efficiently.
- Schedule staff according to expected demand.
- Forecast future sales more accurately.
- Avoid stock shortages or excess inventory.

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5. Would You Describe the Order Data as Highly or Moderately Variable?

Explanation

The data has:

- Mean = 25 orders
- Standard Deviation = 4.77 orders
- Coefficient of Variation = 19.08%

Since the coefficient of variation is less than 20%, the variability is considered moderate to low.

The daily order counts are fairly close to the average, with no extreme fluctuations.

Therefore,

$$\boxed{\text{The order data is moderately variable.}}$$

Conclusion

This case study analyzes the daily order data of an online store using statistical measures. The mean number of orders is 25, the median is 24.5, and the mode is 22, indicating that the data is centered around the mid-20s. The variance (22.8) and standard deviation (4.77) show moderate variation in daily orders. The coefficient of variation (19.08%) confirms that the fluctuations are relatively small compared to the average. Therefore, the store experiences moderately variable demand, making it easier to manage inventory, staffing, and business planning.

Final Answers

Measure	Answer
Mean	25 orders
Median	24.5 orders
Mode	22 orders

Measure	Answer
Variance	22.8
Standard Deviation	4.77 orders

Coefficient of Variation	19.08%
Standard Deviation Indicates	Daily orders vary by about 4.77 orders from the average, showing fairly consistent demand.
Nature of Variability	Moderately variable, as indicated by the standard deviation and coefficient of variation.

Case Study 5 – Comparing Two Data Science Models

Problem Statement

Two machine-learning models have the following prediction errors:

Model A

$$2,\ 3,\ 3,\ 4,\ 4,\ 5,\ 5,\ 6$$

Model B

$$1,\ 1,\ 2,\ 4,\ 5,\ 7,\ 8,\ 9$$

The objective is to compare the performance of the two models using mean, variance, and standard deviation, determine which model is more consistent, and explain whether a lower mean error alone is enough to conclude that one model is better.

Given Data

Model A

$$2,\ 3,\ 3,\ 4,\ 4,\ 5,\ 5,\ 6$$

Number of observations:

$$n = 8$$

Sum of observations:

$$2 + 3 + 3 + 4 + 4 + 5 + 5 + 6 = 32$$

Model B

$$1,\ 1,\ 2,\ 4,\ 5,\ 7,\ 8,\ 9$$

Number of observations:

$$n = 8$$

Sum of observations:

$$1 + 1 + 2 + 4 + 5 + 7 + 8 + 9 = 37$$

---

1. Calculate the Mean Error for Both Models

Definition

The mean error is the average prediction error made by the model.

Formula

$$\bar{x} = \frac{\sum x}{n}$$

where

- $\bar{x}$= Mean
- $\sum x$= Sum of prediction errors
- n= Number of observations

Model A

Calculation

$$\bar{x} = \frac{32}{8}$$

$$\boxed{\bar{x}} \boxed{= 4}$$

Interpretation

The average prediction error of Model A is 4.

Model B

Calculation

$$\bar{x} = \frac{37}{8}$$

$$= 4.625$$

$$\boxed{\bar{x}} \boxed{= 4.625}$$

Interpretation

The average prediction error of Model B is 4.625.

---

2. Calculate the Variance for Both Models

Since all prediction errors are considered, use the population variance formula.

Formula

$$\sigma = \frac{(x - \overline{x})}{n}$$

Model A

Mean

$$\overline{x} = 4$$

Calculation Table

Error (x)	$x - \overline{x}$	$(x - \overline{x})$
2	-2	4
3	-1	1
3	-1	1
4	0	0
4	0	0
5	1	1
5	1	1
6	2	4

$$(x - \overline{x}) = 12$$

Variance

$$\sigma = \frac{12}{8}$$

$$\boxed{\sigma = 1.5}$$

Model B

Mean

$$\bar{x} = 4.625$$

Calculation Table

Error (x)	$x - \bar{x}$	$(x - \bar{x})$
1	-3.625	13.141
1	-3.625	13.141
2	-2.625	6.891
4	-0.625	0.391
5	0.375	0.141
7	2.375	5.641
8	3.375	11.391
9	4.375	19.141

$$(x - \bar{x}) = 70.75$$

Variance

$$\sigma = \frac{70.75}{8}$$

$$\boxed{\sigma = 8.84}$$

---

3. Calculate the Standard Deviation for Both Models

Definition

Standard deviation is the square root of the variance and measures how spread out the prediction errors are around the mean.

Formula

$$\sigma = \sqrt{\sigma}$$

Model A

Calculation

$$\sigma = \sqrt{1.5}$$

$$\boxed{\sigma = 1.22}$$

Interpretation

The prediction errors of Model A are close to the average error, indicating consistent performance.

Model B

Calculation

$$\sigma = \sqrt{8.84}$$

$$\boxed{\sigma = 2.97}$$

Interpretation

The prediction errors of Model B vary much more widely from the average error, indicating less consistent performance.

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4. Which Model Has More Consistent Predictions?

Explanation

Consistency is determined by the standard deviation.

Model	Standard Deviation
A	1.22
B	2.97

A smaller standard deviation means the prediction errors are clustered closer to the mean.

Since

$$1.22 < 2.97$$

$$\boxed{\text{Model A has more consistent predictions.}}$$

Interpretation

Model A produces prediction errors that are more stable and predictable than Model B.

---

5. Can the Model with the Lower Mean Error Necessarily Be Considered Better? Explain

Explanation

No. A model with a lower mean error is not always the better model.

When comparing machine-learning models, several factors should be considered:

- Mean prediction error
- Variance
- Standard deviation
- Consistency of predictions
- Accuracy
- Precision
- Recall
- F1-score (for classification problems)

A model may have a slightly lower average error but very large variations in its predictions, making it unreliable.

In this case:

- Model A
  - Mean Error = 4
  - Standard Deviation = 1.22
- Model B
  - Mean Error = 4.625
  - Standard Deviation = 2.97

Model A has both a lower average error and a much smaller standard deviation, making it the better and more reliable model.

However, in general, a lower mean error alone is not sufficient to conclude that a model is better; consistency and other evaluation metrics must also be considered.

Conclusion

This case study compares two machine-learning models using statistical measures of central tendency and dispersion. Model A has a mean prediction error of 4, while Model B has a mean prediction error of 4.625. The variance (1.5) and standard deviation (1.22) of Model A are much smaller than those of Model B (variance = 8.84, standard deviation = 2.97), indicating that Model A produces more consistent predictions. Therefore, Model A is the more reliable model. This analysis also highlights that a lower mean error alone should not determine the better model; consistency and other performance metrics should also be considered.

Final Answers

Measure	Model A	Model B
Mean Error	4.00	4.625
Variance	1.50	8.84
Standard Deviation	1.22	2.97
More Consistent Model	Model A	—
Is Lower Mean Error Always Better?	No. A lower mean error alone does not guarantee a better model. Consistency (variance and standard deviation) and other evaluation metrics such as accuracy, precision, recall, and F1-score should also be considered.	